CC-S-00795B (DSA-DM)
May 29, 1973
SUPERSEDING
Revision of
Int. Fed. Spec.
CC-S-00795A (DSA-DM)
April 22, 1963

INTERIM FEDERAL SPECIFICATION

SUCTION AND PRESSURE APPARATUS, SURGICAL, EXPLOSION-PROOF, DOUBLE COMPRESSOR, MOBILE, 115 VOLT; 60 CYCLE, AC (With Ether Vaporizer)

This interim Federal specification was developed by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized the use of this interim Federal specification as a valid exception to Federal Specification CC-S-795.

1. SCOPE

1.1 This specification covers one type of explosion-proof suction and pressure apparatus incorporating an ether vaporizer.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

FSC 6515

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SPECIFICATIONS

Federal

CC-M-636	Motor, Alternating-Current, (Fractional Horsepower).
FF-C-77	Casters; Institutional Duty.
QQ-S-698	Steel, Sheet and Strip, Low-Carbon.
TT-E-491	Enamel; Gloss, Synthetic (for Metal and Wood Hospital Furniture).
TT-P-659	Primer-Surfacer; Synthetic, Tints and White, (for Metal and Wood Surfaces).
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-1055	Barrier Material, Waterproofed, Flexible.

Military

MIL-M-7298	Manual, Technical, Commercial Equipment
MIL-P-16232	Phosphate Coatings, Heavy, Manganese or Zinc Base.

STANDARDS

Federal

FED-STD. 66	Steel: Chemical Composition and Hardenability.
FED-STD. 123	Marking for Domestic Shipment (Civilian Agencies).
FED. TEST METHOD STD. NO. 151	Metals, Test Methods
FED-STD-595	Colors
FED. TEST METHOD STD. NO. 601	Rubber: Sampling and Testing.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

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(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402).

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.).

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies).

(Copies of Military specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

AMERICAN SOCIETY FOR TESTING MATERIALS

ASTM Designation C147 — Method of Internal Pressure Test on Glass Containers.

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia, Pa. 19103).

NATIONAL FIRE PROTECTION ASSOCIATION

NFPA No. 56A — Standard for the Use of Inhalation Anesthetics.

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Boston, Mass. 02110).

UNDERWRITERS" LABORATORIES, INC.

UL674(c) — Standard for Electric Motors and Generators for Use in Hazardous Locations, Class I, Groups C and D. Part II - Fractional Horsepower. Motors.

UL894 — Standard for Snap Switches for Use in Hazardous Locations.

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(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 1285 Walt Whitman Road, Melville, Long Island, New York, 11746; 207 East Ohio Street, Chicago, Illinois 60611; or 1655 Scott Boulevard, Santa Clara, California, 95050).

3. REQUIREMENTS

3.1 Fire and casualty hazards.

3.1.1 The contractor shall submit proof, prior to delivery, that the items to be delivered under the contract conform to the applicable requirements of UL 674(c), 894, and NFPA 56A. Proof shall consist of one of the following:

A. UL label or UL listing of the product, provided the contractor agrees to have Underwriters' Laboratories, Inc. forward a copy of the UL report directly to the Contracting Officer, upon the latter's request.

B. Laboratory report from a qualified independent laboratory. If an independent laboratory is used, it must be adequately equipped and competent to determine compliance with UL standards. The qualification of such laboratory shall be determined by the Government which reserves the right to require a facilities report and to conduct a facilities survey. Any laboratory report submitted shall include actual test methods, data taken, and results of tests. Detailed findings of examinations necessary to determine compliance shall also be included. The test report shall also have a section that describes the test sample in such a way that the report can be used to insure identity of production items with the test sample. This section shall be supplemented with construction information such as materials used, spacings between electrical components, types of bushings, photographs, wiring diagrams, sketches, and parts list, all appropriately identified. Compliance with this paragraph does not excuse compliance with all other requirements of the contract and specifications.

3.2.1 Carbon steel. Carbon steel shall conform to QQ-S-698 for C1008, C1010, or C1020 commercial-quality, cold-rolled sheet.

3.2.2 Corrosion-resisting steel. The corrosion-resisting steel shall conform to AISI 304 of FED-STD-66.

3.2.3 Rubber tubing. Rubber tubing shall be electrically conductive and shall have an inside diameter of 1/4 inch, and a 1/8-inch wall. The tubing shall have the following physical properties when tested as specified in 4.4.1.2.1:

Tensile strength (pounds per square inch (psi) minimum)	1200
Tensile strength (psi minimum) after aging	900
Ultimate elongation (percent, minimum)	500

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The tubing shall be electrically conductive, and shall conform to the requirements for Reduction of Electrostatic Hazard - Conductive Accessories, of NFPA No. 56A, when tested as specified in 4.4.1.2.2. In addition, the electrical resistance of the tubing shall be measured after exposure to oxygen aging, liquid ethyl ether, bending and flexing, and oil, as specified in 4.4.1.2.3. Any value in excess of 1 megohm shall be cause for rejection. Proof of acceptability of tubing shall consist of either a listing of the end item by Underwriters' Laboratories, Inc., or a laboratory report from a recognized, qualified laboratory (see 3.1).

3.2.4 Paint. The primer surfacer shall conform to the requirements of TT-P-659. The enamel shall conform to class B of TT-E-491, except that phthalocyanine pigments may be used to obtain the specified color. The color of the enamel shall be light green, conforming to color 14241 of FED-STD-595.

3.2.5 Glass. Glass shall be clear, and annealed.

3.3 Construction. The suction and pressure apparatus shall consist of a cabinet, a motor, two pumps, an anesthesia unit, a suction unit, and filtering devices, all properly connected. The unit shall be provided with the accessories specified in 3.7.

3.3.1 Cabinet. The cabinet shall be of the table type, having a ventilated, sound-deadened compartment. The compartment shall house the motor and compressor. The cabinet shall be provided with handles at the ends and shall have a drawer for the accessories. The overall dimensions of the cabinet shall be as follows:

TABLE I

Width (inches)		Depth (inches)		Height (inches)	
Maximum (Including handles)	Minimum (Excluding handles)	Maximum	Minimum	Maximum	Minimum
32	18	17	12	35	29

Each cabinet shall be provided with four electrically conductive rubber casters, conforming to FF-C-77 for type II, class A not less than size 3 (4-inch diameter wheel shall be acceptable provided it meets minimum requirements for load rating for size 3), style b, 7/16 inch diameter, 1-1/2 inch long stem with retaining band. Two of the casters shall have a locking device. A non-marking, electrically conductive rubber bumper shall surround the lower periphery of the cabinet and shall be firmly secured thereto. The bumper shall extend beyond the outermost projection on the cabinet. The cabinet with casters in place shall conform to the requirements for Reduction of Electrostatic Hazard - Furniture, No. 56A, when tested as specified in 4.4.3.1. The cabinet top shall be of corrosion-resisting steel. As acceptable alternates, the cabinet shall be entirely of corrosion-resisting steel (CRES); or shall be CRES with a base of zinc-coated steel having the manufacturer's standard finish for zinc-coated steel construction.

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3.3.2 Motor. The motor shall conform to CC-M-636 for type I, class 1b, style B4, continuous duty, 1725 revolutions per minute (r.p.m.), 50° centigrade (C.) ambient. The motor shall have not less than 1/4 horsepower and shall operate on 115 volts, 60 cycles alternating current. Proof of acceptability of the motor shall consist of either a listing of the end item by Underwriters' Laboratories, Inc., or a laboratory report indicating compliance with U.L. Standard (see 3.1.1.B). The motor shall be securely mounted within the cabinet in a manner to eliminate vibration and noise. The motor shall be operated by an easily accessible, explosion-proof switch which shall be marked to indicate the "off" and "on" positions. The motor shall be provided with a thermal overload protection. Both the motor and the switch, whether supplied as one unit or as separate components, shall be explosion-proof in accordance with 3.1.

3.3.3 Pumps. Each unit shall be provided with two pumps of the eccentric-rotor rotary-vane type. The pumps shall be either directly connected to the motor shaft or connected through a flexible connection in a manner to insure quiet operation. One pump shall be arranged to create a positive pressure, and the other shall be arranged to create a negative pressure. The pumps shall be provided with means for lubrication and shall not require lubrication more often than every 10 hours of operation.

3.3.4 Anesthesia-unit assembly. The anesthesia-unit assembly shall consist of an ether bottle and valves, a pressure gage, pressure-filter jar, and filter, an ether tip, and the necessary tubing to connect the apparatus. The anesthesia unit shall be so designed that it shall be capable of delivering amounts of ether vapor to the ether tip as indicated below, when tested as specified in 4.4.3.2.

TABLE II

Flow in liters per minute at outlet of ether tip	Ounces (minimum) of ether evaporated after 4 minutes flow time
4	0.220
8	0.376
10	0.450
12	0.528
13	0.559

3.3.4.1 Ether bottles and valves. The ether bottle shall have a capacity of 32 ounces and shall have a pressure relief device incorporated in the positive pressure system. The pressure relief device shall prevent the pressure within the ether bottle from exceeding 11 pounds per square inch (p.s.i.). The pressure relief device shall reset itself automatically. The bottle shall be connected to the positive pressure pump and shall be securely mounted on the top of the cabinet in a manner to permit it to be easily removed. The flow of air through the ether bottle shall be controlled by needle valves or other devices to permit fine adjustments and accurate control of flow from full open to full closed. The valves or devices shall be marked or calibrated to indicate their position relative to full open or full closed. The ether bottle

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and pressure filter jar shall be of such construction as to withstand the internal pressure test specified in 4.4.2.2.1 without failure. Bottles and jars subjected to this pressure test shall be discarded.

* 3.3.4.2 Pressure gage. The anesthesia unit shall be provided with a pressure gage calibrated to indicate pressures from 0 to 30 p.s.i., in increments of one p.s.i. This gage shall be so connected in the anesthesia system as to show the pressure within the pressure system. The gage shall be accurate to within ± 2 percent over the center half of the scale, and ± 3 percent over the remainder of the scale.

3.3.4.3 Anesthesia-system filters. A series of filters shall be provided in the anesthesia system to prevent dust particles and oil from entering the air stream. The filtering medium shall be easily replaceable.

3.3.5 Suction assembly. The suction assembly shall consist of a suction bottle, suction filter bottle valves, a gage, and filter.

* 3.3.5.1 Suction bottles. The collection bottle shall be 1/2 to 1 gallon in capacity. The bottle shall be securely mounted on top of the cabinet in a manner to allow easy removal. The bottle shall be graduated in both ounces, and in the metric system of measurement. The suction bottle and suction filter jar shall pass the vacuum test in 4.4.2.2.2.

* 3.3.5.2 Suction-system gage. The suction unit shall be provided with a suction gage calibrated to indicate negative pressures from 0 to 30 inches of mercury, in increments of 1 inch. This gage shall be connected to the suction system to show the negative pressure in the suction bottle. The gage shall be accurate to within ± 2 percent over the center half of the scale, and ± 3 percent over the remainder of the scale.

3.3.5.3 Suction-system filters. Filters shall be provided in the suction system, interposed between the suction bottle and the suction pump, to protect the pump from foreign matter. The filtering medium shall be easily replaceable. In addition, a device shall be provided which shall cut off the negative pressure from the suction bottle before it starts to overflow.

3.4 Rubber tubing and couplings. Each unit shall be provided with sufficient tubing to perform its intended function. The tubing shall be provided with couplings fabricated of corrosion-resisting steel, or of brass or steel, plated with chromium over nickel. The couplings shall be of a tapered slip-type, with a positive screw lock. The tapered portion of the couplings shall be tapered for a length of 0.250 to 0.275, ± 0.005 from a diameter of 0.351 ± 0.005 inch to a diameter of 0.250 ± 0.005 inch. The female portion of the coupling shall have a corresponding taper. The male part of the coupling shall be provided with a threaded portion, 7/16 inch by 27 NC thread, which shall mate with a threaded collar on the female part of the coupling.

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The collar and threaded portion shall insure locking of the coupling to form a tight, leakproof connection. As an alternate, 17/64 inch I.D. lumen "o" ring seal, leak-proof fittings may be supplied on the collection bottle, consisting of male and threaded collar fittings on pump side, and female fitting on the patient side.

4 3.5 Performance. The delivery at the pressure end of the tubing, with valves full open, shall be at least 1/2 cubic foot per minute. Shall be capable of delivering a minimum positive pressure of 20 psig to the pressure filter bottle, and a minimum vacuum of 20 inches of mercury to the collection bottle. Bottles and jars shall pass the test in 4.4.3.5.

3.6 Electric wiring. Electrical plugs and receptacles within the unit shall be explosion-proof, approved for use in Class I, Group C locations. The unit shall be equipped with a flexible, insulated, three-wire cord, not less than 10 feet long. The third or ground wire shall be properly identified. The cord shall not be furnished with a male plug, but shall bear a red warning tag at the free end cautioning that only an explosion-proof attachment plug (to mate with the hospital explosion-proof receptacle) shall be installed by the using activity. The unit shall withstand the dielectric strength test specified in 4.4.3.3.

3.7 Accessories. Each suction and pressure apparatus shall be furnished with one each of the following accessories (see 6.3):

Yankauer laryngeal cannula

Poole abdominal cannula

Ether hook.

Each accessory shall be fitted with a male taper-slip connection to mate with the hose coupling on the tubing (see 3.4).

3.8 Finish.

3.8.1 Cabinet. All interior and exterior surfaces of the cabinet, except the top, shall be finished as follows:

3.8.1.1 Cleaning and preparation. Before finishing, parts to be painted shall be cleaned with suitable solvents and phosphate treated in accordance with type M, class 1 of MIL-P-16232, or other equivalent chemicals to insure rust resistance of the metal and to provide a satisfactory bond for the finishing materials. All castings shall be polished to remove sharp edges prior to surface treatment and application of primer.

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3.8.1.2 Primer. After cleaning and preparation, the parts shall be given one coat of a synthetic primer conforming to TT-P-659. The primer shall be baked on for not less than 45 minutes, at a temperature of 250° Fahrenheit (F.), or for an equivalent baking period at a higher or lower temperature as recommended by the paint manufacturer. The primer shall be applied so as to level out and produce a uniform surface, free of pebbles and wrinkles.

3.8.1.3 Enamel. After the primer has been allowed to cool, all surfaces shall be covered with two coats of synthetic gloss enamel conforming to class B of TT-E-491, except that phthalocyanine pigments shall be used to obtain the specified color. Each coat shall be baked on separately for not less than 45 minutes at a temperature 250°F. or for an equivalent baking period at a higher or lower temperature as recommended by the paint manufacturer. The enamel coats shall be applied so as to level out and produce a final, smooth, uniform surface without runs, wrinkles, orangepeel, or grit, and without areas of thin or no film. The total dry-film thickness shall be not less than 0.002 inch at any point.

3.8.2 Cabinet top. The corrosion-resisting steel cabinet top shall be given a uniform, No. 4 commercial finish.

3.8.3 Hardware. Hardware, including the fittings and fixtures, shall be chromium plated and polished to a mirror finish; or corrosion-resisting steel polished to a No. 4 commercial finish.

* 3.9 Identification marking. Each unit shall be permanently and legibly marked with the National Stock Number, name or registered trademark of the manufacturer, oiling instructions, the contract or purchase order number, the letters U. S., item identification, the serial number, voltage, amperage, frequency rating, and such other data as may be required by Federal laws. In addition, the motor shall be marked in accordance with CC-M-636. Marking secured by adhesives shall not be used. Markings, except motor marking, shall be clearly legible and located to allow its being visible during use.

* 3.10 Service data. The contractor shall furnish, with each unit, two copies each of a technical manual which contains complete instructions for installation, operation, and maintenance and lists of component parts. The service data shall include a description of the pressure relief device and all necessary instructions and precautions concerning its operation. The manual shall comply with the requirements of MIL-M-7298. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the Contracting Officer, via the cognizant Government inspector, two preliminary copies of his proposed technical manual, for review and comment by the Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary manuals shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor and any required revisions and corrections in the final manual without delaying delivery under the con-

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tract or purchase order. In addition, the Contractor shall furnish 15 copies of the approved technical manual to the Contracting Officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of contract and propose to furnish manuals identical with those previously accepted, then the requirements for review of drafts and submission of 15 additional copies may be waived upon request.

3.11 Center of gravity and net weight information. The contractor shall prepare a drawing, figure, or illustration, indicating the position of the vertical center of gravity of the unit with reference to its base. The net weight of the unit shall also be shown. In the event the required information has been submitted previously, no further report is necessary unless changes in the unit involve a variation greater than ten percent in center of gravity or net weight. Four copies shall be submitted by the contractor directly to Chief, Technical Services Branch, Attention: Code DPSC-ATTP, Defense Personnel Support Center, Directorate of Medical Material, 2800 South 20th Street, Philadelphia, Pa. 19101.

3.12 Welding. Welding shall be durable, sound in every detail and in accordance with latest practices recommended by the American Welding Society. All welds shall be produced by qualified welding operators with satisfactory welding equipment using satisfactory filler metal and/or electrode, where applicable. Surfaces to be welded shall be free from grease, oil, oxides or any other foreign matter. Any cleaning method which will remove these substances and leave the surface clean is satisfactory. All aluminum parts shall be welded within 48 hours after a chemical cleaning treatment. Parts must be chemically recleaned if the 48 hour time period is exceeded. All weld joints shall blend into the adjacent metal in gradual, smooth curves. Beads shall be smooth and free of slag, excessive undercut, or excessive spatter. Sufficient metal shall be added to provide a suitable fillet or reinforcement unless otherwise specified. Fusion through the full thickness of the base metal, in areas other than those where 100% penetration is a requirement of joint design, is not acceptable when welding materials over 0.050 inch in thickness. Welds shall not be oxidized. All welds shall be continuous with no visible voids, pin holes, or cracks.

3.13 Workmanship. Workmanship shall be first class throughout. The suction and pressure apparatus shall be free from any imperfections which affect its appearance or which may impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, and auditory, investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

* 4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the raw materials may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the National Stock Number, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and Table III.

TABLE III

	Inspection level	AQL (defects per hundred units)
For visual examination		
Major defects	II	1.0
Minor defects	II	4.0
Dimensional examination	S-3	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and Table IV. AQL shall be zero for all tests; but in the event of a failure during 100 percent inspection; the entire lot shall not be re-tested.

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TABLE IV

Component (unit of product)	Characteristic	Requirement	Test procedure	Inspection level
Metals	Chemical composition	3.2.1, 3.2.2	4.4.1.1	(1)
Rubber tubing	Physical properties	3.2.3	4.4.1.2.1	(2)
	Electrical conductivity	3.2.3	4.4.1.2.2	(2)
	Electrical resistance	3.2.3	4.4.1.2.3	(2)
Safety valve	Relief pressure	3.3.4.1	4.4.2.1	100 percent
Ether and pressure filter bottle	Internal pressure	3.3.4.1	4.4.2.2.1	S-1
Pressure and Vacuum gages	Accuracy	3.3.4.2 and 3.3.5.2	4.4.2.3	S-1
Suction and suction filter bottle	Vacuum	3.3.5.1	4.4.2.2.2	S-1
End item	Electrical conductivity	3.3.1	4.4.3.1	100 percent
	Ether vaporization	3.3.4	4.4.3.2	(3)
	Suction overflow	3.3.5.3	4.4.3.4	S-1
	Performance	3.5	4.4.3.5	100 percent
	Dielectric strength	3.6	4.4.3.3	100 percent

(1) Four ounces of each lot of metal used in the construction of the apparatus shall be selected for chemical analysis.

(2) Two samples of rubber tubing representative of each lot used in the manufacture of the apparatus shall be selected for test. Each sample shall be large enough to provide the required number of specimens for the tensile tests, both before and after aging, and for the electrical resistance tests. In addition, two 6-inch lengths of the intact tubing shall be taken, one for aging, and one as a control.

(3) Two units shall be tested. Provided design of unit has not been changed, this test need not be repeated by a manufacturer who has successfully completed the test under a previous contract.

4.3 Examination. The suction and pressure apparatus shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects:

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Classification of defects*	
Major	
101	Unit not complete with all components or accessories.
102	Exposed edges and corners not free of burs or sharpness.
103	Any weld not fused; not free of burns, cracks, or fractures.
104	Any threaded part stripped.
105	Doors not free of warp or deformity.
106	Unit not clean; not free of dirt, oil, or grease.
107	Sound-deadening material not supplied.
108	Any bottle not free of cracks, breaks, chips, sharp or deformed areas.
109	Tapered slip-type couplings do not mate properly.
110	Finish not free of blistering, checking, peeling.
111	Identification marking not present, not complete, not legible, not properly attached.
Minor	
201	Drawer does not operate smoothly.
202	Drawer does not close flush with cabinet.
203	Rubber bumper not securely held in place.
204	Bottle not stable (rocks on flat surface).
205	Any screw not drawn up tight; head not flush with the surface held.
206	Color of enamel finish not as specified.
207	Service data not included.
208	Paint finish not free of tackiness, softness or sags.
209	Paint finish not uniform; not free of imbedded foreign matter.
210	Finish not free of scratches, bare spots, or areas of visibly thin coating.

* Examination shall not be restricted to the classified possible defects tested above.

4.4 Tests. Tests shall be conducted, as necessary, to determine compliance with specification requirements.

4.4.1 Materials.

4.4.1.1 Steel.

4.4.1.1.1 Chemical composition. Tests for chemical analysis shall be conducted in accordance with method 111.1 or 112.1 of FED-STD-151.

4.4.1.2 Rubber tubing.

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4.4.1.2.1 Physical properties. Physical properties shall be determined in accordance with the appropriate test method of FED-STD-601.

Property	Test method
Tensile strength	4111
Ultimate elongation	4121
Oxygen pressure	7111*
Air heat	7221*

* Only one of these tests is required.

4.4.1.2.2 Electrical conductivity. The rubber tubing shall be tested for electrical conductivity in accordance with the applicable procedures outlined in NFPA No. 56A.

4.4.1.2.3 Electrical resistance. The electrical resistance of the rubber tubing shall be determined after being subjected to the following tests.

4.4.1.2.3.1 Oxygen aging. Two 7-inch lengths of tubing shall be subjected to oxygen under pressure of 300 p.s.i. for a period of 96 hours at a temperature of 70°C. The electrical resistance of the tubing shall be measured before and after the oxygen aging.

4.4.1.2.3.2 Liquid ethyl ether. Two 5-foot lengths of tubing shall be maintained for 5 hours full of ethyl ether in a liquid phase. The electrical resistance of the tubing shall be measured initially and at 1/2-hour intervals during the test.

4.4.1.2.3.3 Bending and flexing. Two 5-foot lengths of tubing fitted with electrical test connections, shall be passed back and forth for 1 hour a minimum of 10 times per minute on a pulley 1-3/4 inches in diameter. The tubing shall be held in tension by a 5-pound weight suspended from the free end of the sample. The electrical resistance of the tubing shall be measured initially and at 20 minute intervals during the test and at completion of test.

4.4.1.2.3.4 Oil. Two 7-inch lengths of tubing shall be submerged for 5 days in SAE 30 motor oil, Type SA. The resistance of the tubing shall be measured before and after the test.

4.4.2 Components.

4.4.2.1 Safety valve. The safety valve shall be tested in the manner in which it is mounted on the unit. The pressure on the safety valve shall be increased gradually and the maximum pressure value attainable shall be recorded. The safety valve shall also be examined to determine whether or not it has re-set itself after relief.

* 4.4.2.2 Bottles.

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4.4.2.2.1 Pressure bottles. The bottle and pressure filter jar shall be subjected to an internal pressure of 45 + 2 -0 psig for a minimum of 1 minute. Air or liquid pressure shall be acceptable. For liquid pressure, the test procedures shall be in accordance with ASTM C147. If air pressure is used, precaution should be taken to contain possible glass breakage. After testing, bottles and jars shall be discarded.

4.4.2.2.2 Suction bottles. The suction bottle and suction filter jar shall be subjected to an internal vacuum of 25 inches of mercury for a period of 1 minute without failure.

4.4.2.3 Gages. Gages shall be examined to determine that the calibrations conform to requirements. The gages shall be measured for accuracy using standard gages accurate to the nearest 1/2 percent.

4.4.3 End item.

4.4.3.1 Electrical conductivity. The apparatus, complete with casters and rubber tubing, shall be tested in accordance with the applicable procedures outlined in NFPA No. 56A.

* 4.4.3.2 Ether-vaporization rate. The ether bottles shall be filled with U.S.P. grade ethyl-ether to cover the diffusing element of the bubbling tube to a height of 2 ± 1/16 inch above the top of the diffusing element. The ether bottle, including the contents and the bubbling tube, shall be weighed and then chilled to such a temperature that at the beginning of the test the temperature shall be 32 ± 1°F. (0° ± 1/2°C). The ether bottle shall be assembled to the apparatus and the apparatus operated at the prescribed rates of flow (see 3.3.4) for 4 minutes. At the end of each 4-minute flow test, the ether bottle, contents and bubbling tube shall be removed and weighed. Room conditions during the test shall be 73.4 ± 2°F. (23 ± 1°C.) and 45 ± 5 percent relative humidity. Two determinations at each flow rate for each apparatus tested shall be made and averaged to determine conformance to 3.3.4.

* 4.4.3.3 Dielectric strength. A dielectric test potential of 900 volts, 60 cycles, single phase, alternating current, shall be applied for 1 minute between the exposed non-current-carrying parts and the current-carrying parts, with all current-carrying parts in the circuit. Test shall be performed at the maximum operating temperature reached in normal use.

4.4.3.4 Suction overflow. The suction system shall be operated until the cut-off device for the negative pressure operates. The device shall operate before any liquid from the suction bottle is drawn into the suction system.

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* 4.4.3.5 Performance. The unit shall be operated in accordance with the manufacturer's instructions, and the tests for the following characteristics shall be made to determine conformance to 3.5

Compressor capacity.
Delivery at pressure end.
Intake at suction end.

The suction and ether bottles, and the suction and pressure filter jars shall be subjected to a minimum pressure of 20 psig and a minimum vacuum of 20 inches of mercury, as applicable, for a period of one minute, without damage. Test gages shall have an accuracy of ± 1 percent.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or C as specified (see 6.2).

5.1.1 Unit of issue. One suction and pressure apparatus, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit package. Each unit shall be disassembled as far as is practicable and packaged as follows:

5.1.2.1.1 Disassembled parts. Disassembled parts shall be adequately protected by wrapping with flexible corrugated fiberboard. Parts so wrapped shall then be packaged in a nailed wood box of suitable size and design. The box shall contain sufficient cushioning material (corrugated pads, air cells, or cellulose wadding) to insure against shifting and breakage.

5.1.2.1.2 Motor and pumps. If the motor and pumps are flexibly mounted (floating type), they shall be adequately blocked or supported and secured with steel bands where practicable, in a manner to prevent movement of motor or pump while in transit or storage. A tag shall be attached to the cabinet directing the removal of all shipping straps and blocks prior to use of equipment.

5.1.2.1.3 Cabinet. Doors and drawers of the cabinet shall be secured to prevent accidental opening while in transit or storage. The cabinet shall then be wrapped with kraft paper of suitable weight and padded with corrugated fiberboard or with excelsior pads. Wrapping and padding shall be firmly secured with kraft gummed paper tape, or with cotton twine or tape, or with any combination of the three. All work shall be accomplished in a manner to prevent scratching or marring of finished surfaces or damage to the cabinet.

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5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which shall afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. The cabinet, and the box containing disassembled components, shall be packed in an exterior container designed for type 3 load and constructed in accordance with Federal Specification PPP-B-621, class 2. Box and cabinet shall be adequately secured within the exterior container by means of bolts, wood braces, or other interior fittings in such a manner as to insure against shifting, damage, or marring of surfaces. All box panels, except the base, shall be lined with waterproof barrier material conforming to PPP-B-1055, class M-1 in such a manner as to prevent direct entry of water. When gross weight of the exterior container exceeds 250 pounds, exterior container design shall include skids of not less than 1-1/2 inches in thickness; applied crosswise of the base, so positioned as to facilitate handling with a forklift truck and to maintain unit in an upright position. The cabinet shall be blocked or braced in such a manner that the weight shall not be carried directly by the casters, and any protruding parts of the cabinet shall not contact the inner walls of the container. Closure shall comply with appendix of box specification. Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. The cabinet, and the box containing disassembled components, shall be packed in an exterior container designed for type 3 load and constructed in accordance with Federal Specification PPP-B-621, Class 1. Box and cabinet shall be adequately secured within the exterior container by means of bolts, wood braces, or other interior fittings in such a manner as to insure against shifting, damage, or marring of surfaces. When gross weight of the exterior container exceeds 250 pounds, exterior container design shall include skids of not less than 1-1/2 inches in thickness, applied crosswise of the base, so positioned as to facilitate handling with a forklift truck and to maintain unit in an upright position. The cabinet shall be blocked or braced in such a manner that the weight shall not be carried directly by the casters, and any protruding parts of the cabinet shall not contact the inner walls of the container. Closure shall comply with appendix of box specification. Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

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5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Civil agencies. Interior packages and exterior containers shall be marked in accordance with Federal Standard No. 123.

5.3.2 Military activities.

5.3.2.1 Interior packages. Interior packages shall be marked to show the contents contained therein.

5.3.2.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Marking shall include arrows as specified in MIL-STD-129. Marking shall also include the words "Front" or "Open this Side" as appropriate in order to facilitate opening.

6. NOTES

6.1 Intended use. This suction and pressure apparatus is intended for use during surgical procedures in Class I, Group C hazardous locations.

6.2 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

* 6.3 The Yankauer larynegeal cannula described in 3.7 shall conform to the latest requirements for National stock number 6515-00-721-9474. The Poole abdominal cannula described in 3.7 shall conform to the latest requirements for National stock number 6515-00-721-9475. The ether hook described in 3.7 shall conform to the latest requirements for 6515-00-721-9491.

6.4 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, which are normally procured to meet Federal Government requirements.

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6.5 This specification covers the following item shown in the Federal Supply Catalog.

National Stock Number	Item
6515-00-375-3100	Suction and Pressure Apparatus, Surgical Explosion Proof, Double Compressor, Mobile, 115 Volts, 60 cycle, AC (with Ether Vaporizer)

6.6 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Review activities:
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Air Force - 03

Preparing activity:
Defense Supply Agency - DM

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